# Session 7: More data visualisation and ggplot

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# Overview I

# Introduction

# Learning goals of today

- Short repetition from last week

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- Short repetition from last week
- Some more data viz examples

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# Learning goals of today

- Short repetition from last week
- Some more data viz examples
- Some more ggplot options

## Repetition: Basic components of ggplot() function

- call function: ggplot(...

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- call function: ggplot(...
- add data: data = ...

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# Repetition: Basic components of ggplot() function

- call function: ggplot(...
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- add aesthetics aes(x = ... , y = ... , color = ...)

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# Repetition: Basic components of ggplot() function

- call function: ggplot(...
- add data: data = ...
- add aesthetics aes(x = ... , y = ... , color = ...)
- add layer that you want to visualise, like points + geom_point()

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# Repitition: Simple example of ggplot() - Code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```

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# Other important things to remember

- ggpplot() usually requires long, tidy data

# Other important things to remember

- ggpplot() usually requires long, tidy data
- Meaning information you want to diplay need to be stored in one separate variables, not many separate variables

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# What additional things

- Including a third variable

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# What additional things

- Including a third variable
- Scaling

## What additional things

- Including a third variable
- Scaling
- Labelling

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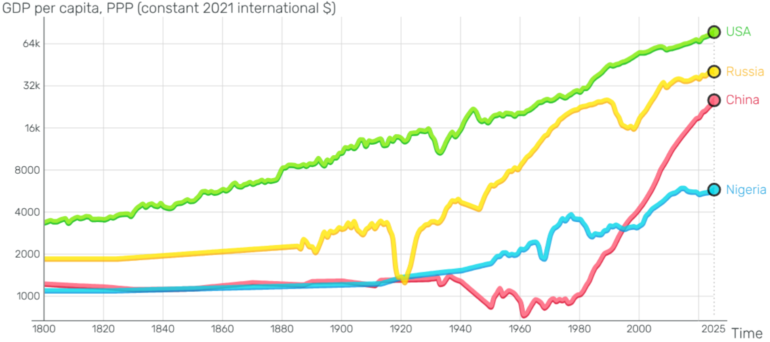
# Third variable

# Great example: Colour




Source: www.gapminder.org

# Options for third variable

- There are many ways to include 3rd variable into graph

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- *colour* / *fill*

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# Options for third variable

- There are many ways to include 3rd variable into graph
- *colour* / *fill*
- *size*

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# Options for third variable

- There are many ways to include 3rd variable into graph
- *colour* / *fill*
- *size*
- *shape*

# Options for third variable

- There are many ways to include 3rd variable into graph
- *colour* / *fill*
- *size*
- *shape*
- *alpha*

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# Options for third variable

- There are many ways to include 3rd variable into graph
- *colour / fill*
- *size*
- *shape*
- *alpha*
- *facet_wrap() / facet_grid()*

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- There are many ways to include 3rd variable into graph
- *colour / fill*
- *size*
- *shape*
- *alpha*
- *facet_wrap() / facet_grid()*

# *colour* / *fill*

- Adding *colour* or *fill* adds colour to graph

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# *colour* / *fill*

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- lines and strokes (e.g. geom_point()/ geom_line()) –> *colour*

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# *colour* / *fill*

- Adding *colour* or *fill* adds colour to graph
- lines and strokes (e.g. geom_point()/ geom_line()) –> *colour*
- inside areas (e.g. geom_bar()/ geom_hist()) –> *fill*

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# *colour* / *fill*

- Adding *colour* or *fill* adds colour to graph
- lines and strokes (e.g. geom_point()/ geom_line()) –> *colour*
- inside areas (e.g. geom_bar()/ geom_hist()) –> *fill*
- Colours depend on input variable (factor vs. continuous)

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# *colour* (factor): Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with colour
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```

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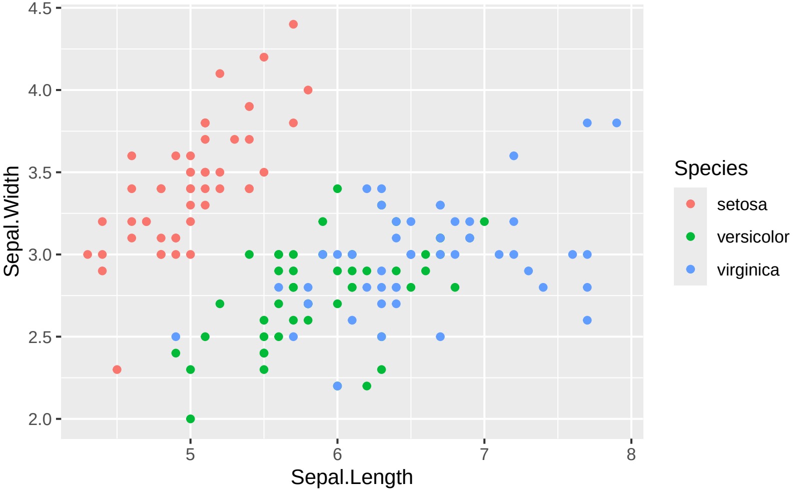
# *colour* (factor): Example image

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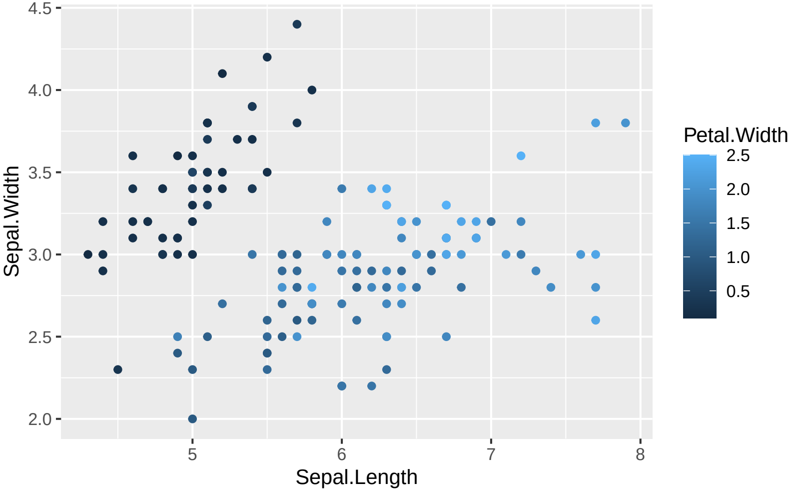
# *colour* (continuous): Example code

```r
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with colour (continuous)
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Petal.Width)) +
  geom_point()
```

# *colour* (continuous): Example image

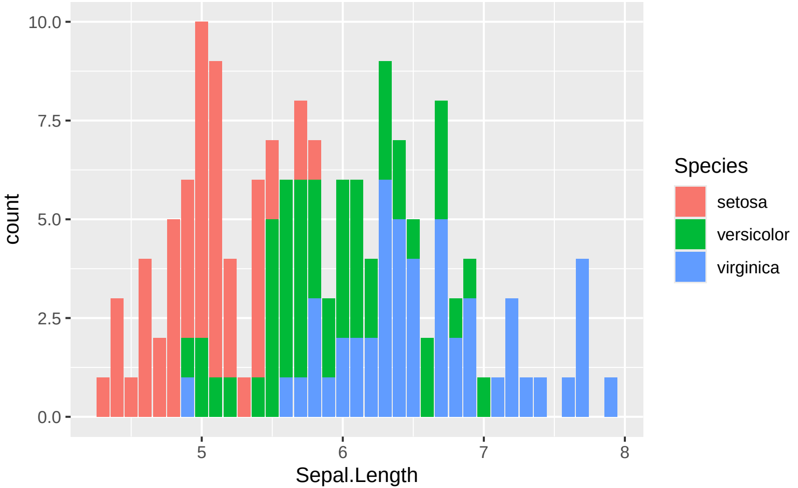


# *fill* (factor): Example code

```r
# Load ggplot2
library(ggplot2)

# Bar plot using iris data set with fill
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           fill = Species)) +
  geom_bar()
```

# *fill* (factor): Example image



# Exercise: Create three colour graphs with your data

- Create one graph using fill (factor)

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# Exercise: Create three colour graphs with your data

- Create one graph using fill (factor)
- Create one graph using colour (factor)

# Exercise: Create three colour graphs with your data

- Create one graph using fill (factor)
- Create one graph using colour (factor)
- Create one graph using fill or colour (continuous)

## *size*

- for geom_point(), adding *size* changes size of points

# *size*

- for geom_point(), adding *size* changes size of points
- Can be useful to highlight larger weights of individual points (e.g. country population)

## *size*: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with size
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           size = Petal.Width)) +
  geom_point()
```

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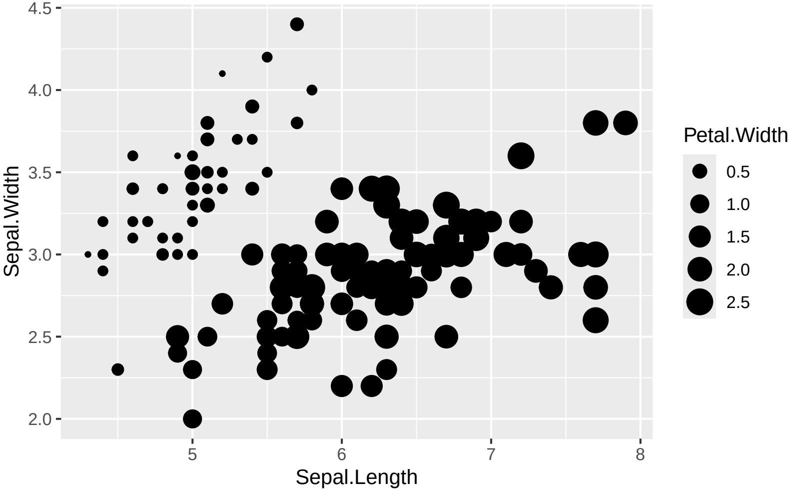
# *size*: Example image



## *shape*

- for geom_point(), adding *shape* changes shape of points

# *shape*

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- Can be useful to visually highlight different sub-divisions in data

# *shape*

- for geom_point(), adding *shape* changes shape of points
- Can be useful to visually highlight different sub-divisions in data
- Originally more useful for black and white applications, nowadays colours used more regularly

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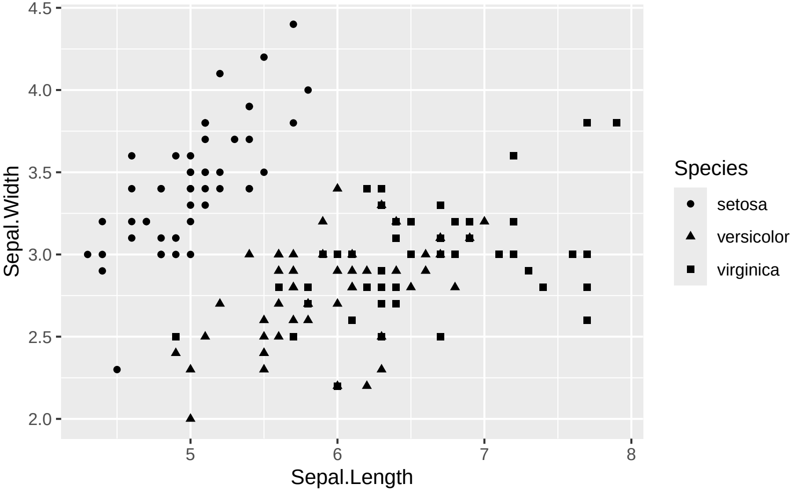
## *shape*: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with shapes
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           shape = Species)) +
  geom_point()
```

# *shape*: Example image

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# Combining options *colour* and *shape*

- Of course, one can also combine options

# Combining options *colour* and *shape*

- Of course, one can also combine options
- may increase visibility of differences even more

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# *colour* + *shape*: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with colour and shapes
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species,
           shape = Species)) +
  geom_point()
```

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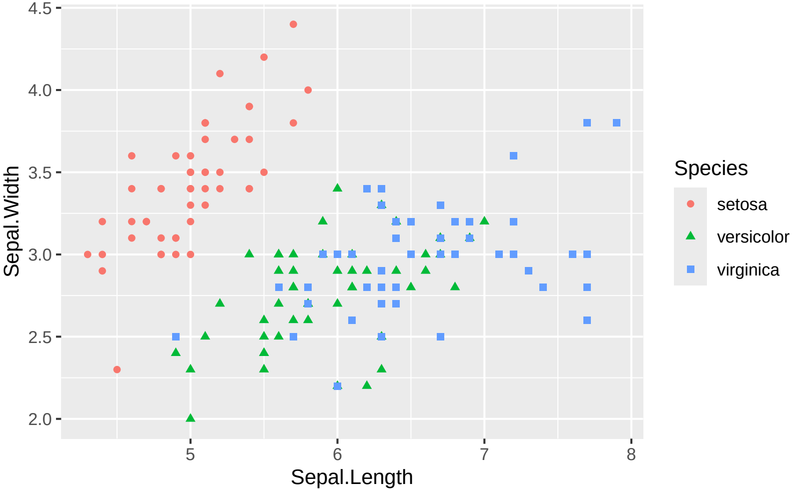
# *colour* + *shape*: Example image

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# *alpha*

- *alpha* sets the opacity of information

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# *alpha*

- *alpha* sets the opacity of information
- Can be useful to visually highlight different sub-divisions in data
- Originally more useful for black and white applications, nowadays colours used more regularly
- Remember the bar plot from before?

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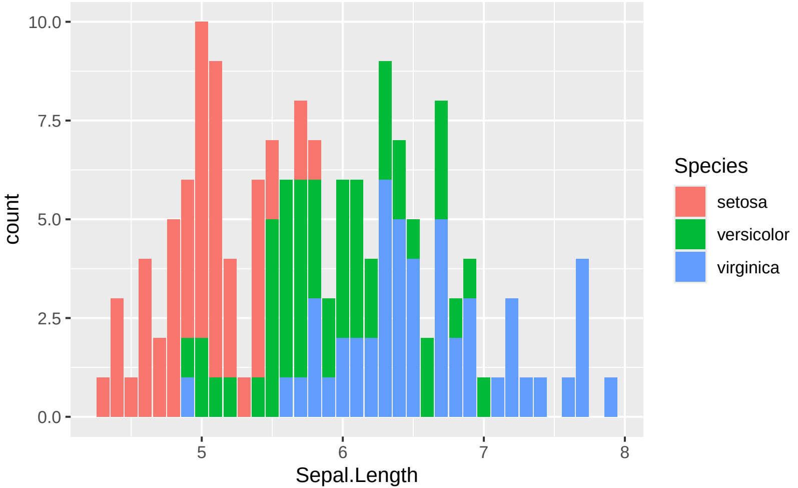
# *fill* (factor): Example code

```
# Load ggplot2
library(ggplot2)

# Bar plot using iris data set with fill
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           fill = Species)) +
  geom_bar()
```

# *fill* (factor): Example image

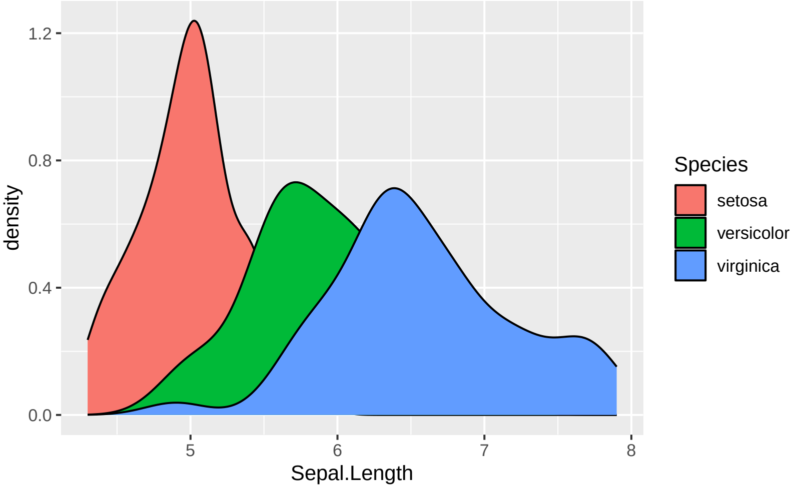


# *fill* (density): Example code

```
# Load ggplot2
library(ggplot2)

# Density plot using iris data set with fill
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           fill = Species)) +
  geom_density()
```

# *fill* (density): Example image

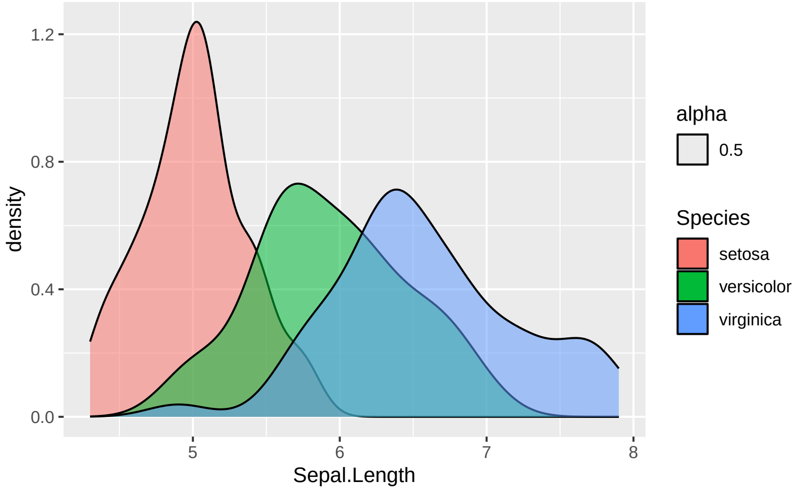


# *fill* (density) + *alpha*: Example code

```
# Load ggplot2
library(ggplot2)

# Density plot using iris data set with fill and alpha
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           fill = Species,
           position = "identity",
           alpha = 0.5)) +
  geom_density()
```

# *fill* (density) + *alpha*: Example image



# *alpha* as variable

- Of course, *alpha* can also be used with variable

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# *alpha* as variable

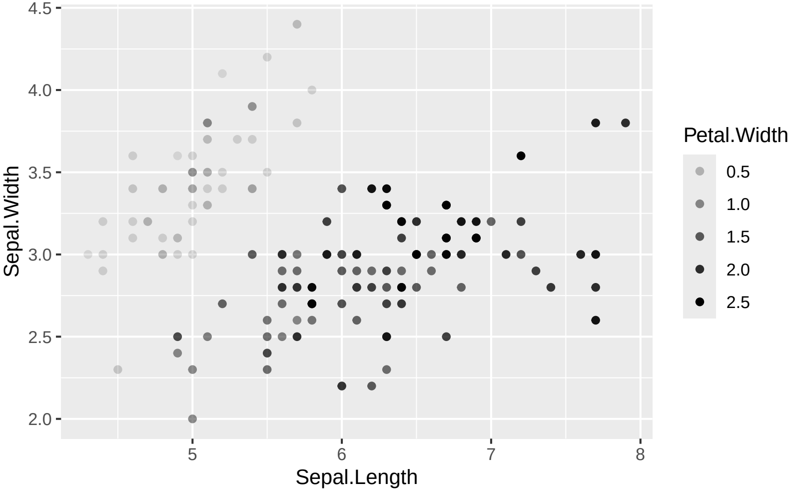
- Of course, *alpha* can also be used with variable
- Though use case less obvious

# *alpha*: Example code

```r
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with shapes
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           alpha = Petal.Width)) +
  geom_point()
```

# *alpha*: Example image



# Exercise: Create three graphs with your data

- Create one graph using *size*

# Exercise: Create three graphs with your data

- Create one graph using *size*
- Create one graph using *shape*and *colour*

# Exercise: Create three graphs with your data

- Create one graph using *size*
- Create one graph using *shape*and *colour*
- Create a density plot with *fill* and *alpha*

# Exercise: Create three graphs with your data

- Create one graph using *size*
- Create one graph using *shape*and *colour*
- Create a density plot with *fill* and *alpha*
- Consider using different variables than in the last exercise

# *alpha*: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with size
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           alpha = Petal.Width)) +
  geom_point()
```

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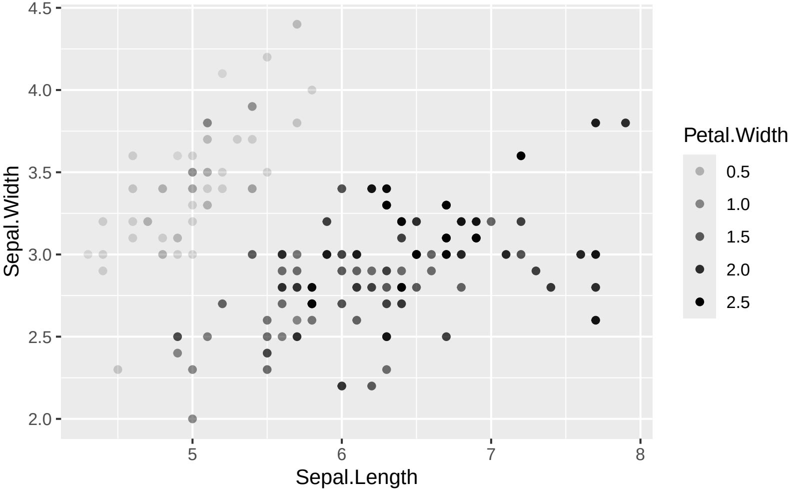
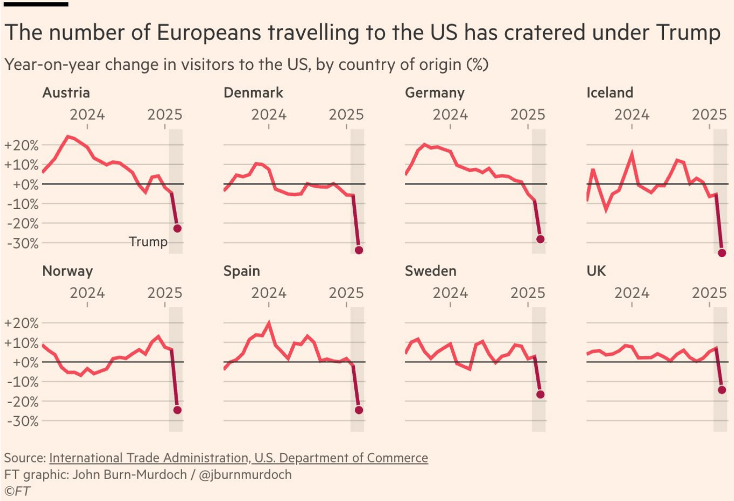
# *alpha*: Example image



# Great example: Faceting

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# *facet_wrap() / face_grid()*

- faceting is super useful, especially with large data sets

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- faceting is super useful, especially with large data sets
- Creates separate graph per sub-group (hence requires factor variable)
- *facet_wrap(~ x)* <- best to use with one factor variable (x), plots separate graphs per factor

# *facet_wrap() / face_grid()*

- faceting is super useful, especially with large data sets
- Creates separate graph per sub-group (hence requires factor variable)
- *facet_wrap(~ x)* <- best to use with one factor variable (x), plots separate graphs per factor
- *facet_grid(x ~ y)* <- best to use with two factor variables (x, y), plots grid of all combinations of x and y

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# *facet_wrap()*: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with facet_wrap()
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width)) +
  geom_point() +
  facet_wrap(~ Species)
```

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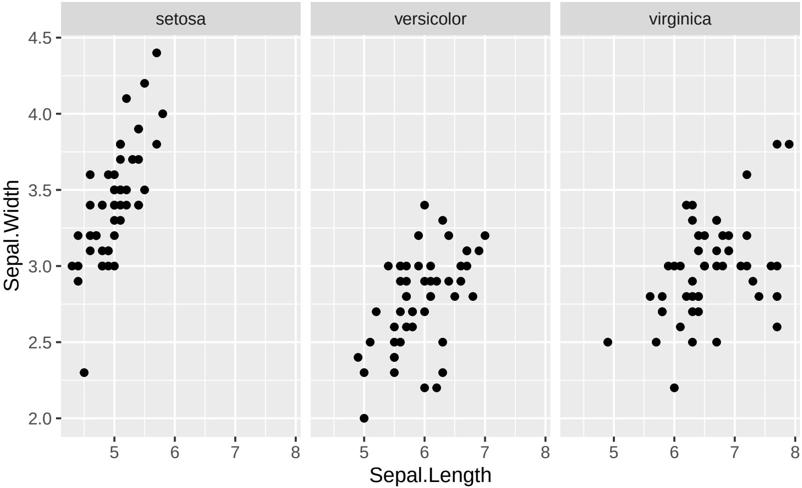
# *facet_wrap()*: Example image



# *facet_grid()*: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using mtcars data set with facet_wrap()
ggplot(data = mtcars,
       mapping =
         aes(x = hp,
           y = mpg)) +
  geom_point() +
  facet_grid(am ~ cyl)
```

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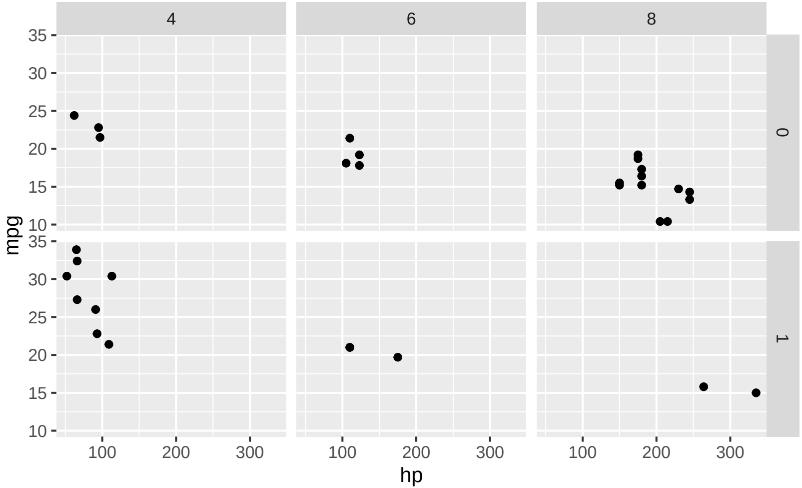
# *facet_grid()*: Example image



# Exercise: Create two graphs with your data

- Create one graph using *facet_wrap*

# Exercise: Create two graphs with your data

- Create one graph using *facet_wrap*
- Create one graph using *facet_grid()*

# Things to keep in mind

- Not all ways to integrate third variable work with every geom

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- Not all ways to integrate third variable work with every geom
- *colour* and *fill* work with nearly all geoms

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# Things to keep in mind

- Not all ways to integrate third variable work with every geom
- *colour* and *fill* work with nearly all geoms
- Otherwise, just try out and see what works (for you)

# Scaling and labelling

# Scaling the axes

- Sometimes graphs are too messy so we want to scale

## Scaling the axes

- Sometimes graphs are too messy so we want to scale
- one can either adapt data in pre-processing

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# Scaling the axes

- Sometimes graphs are too messy so we want to scale
- one can either adapt data in pre-processing
- Or one can limit the axes

# Scaling through pre-processing (before)

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data, but only below Sepal.Length
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```

# Scaling through pre-processing (after)

```
# Load ggplot2
library(ggplot2)

# Load the tidyverse
library(tidyverse)

# Scatterplot using iris data, but only below Sepal.Length
iris |>
  mutate(Sepal.Length = ifelse(Sepal.Length <=7, Sepal.Leng
  ggplot(mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```

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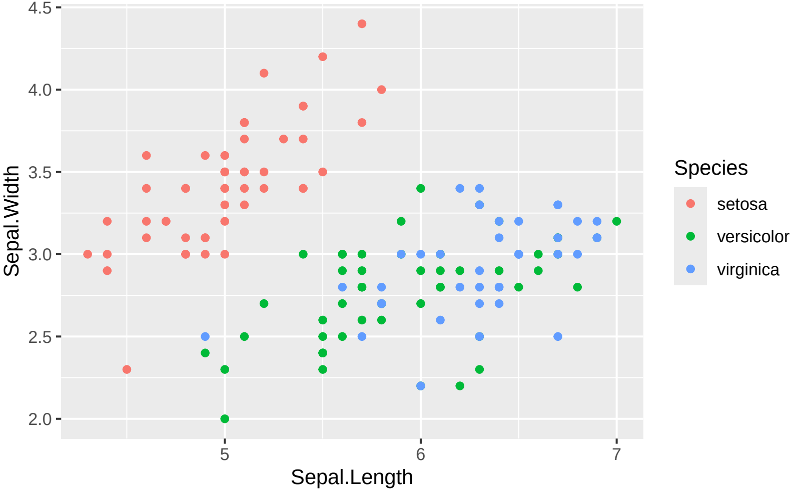
# Scaling through pre-processing (after): Image

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# Pros and Cons of pre-processing

- One actually limits the data “by hand” by oneself

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# Pros and Cons of pre-processing

- One actually limits the data “by hand” by oneself
- Better understanding of data

# Pros and Cons of pre-processing

- One actually limits the data "by hand" by oneself
- Better understanding of data
- May affect some statistics (like means etc.) in graph

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# Alternative: Limiting axes

- Alternative is adding + xlim() or + ylim() to ggplot()

# Alternative: Limiting axes

- Alternative is adding + xlim() or + ylim() to ggplot()
- Both lower and upper are required, but can be set to NA

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# Scaling through xlim() and ylim() (before)

```r
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data, but only
# below Sepal.Length of 7
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```

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# Scaling through pre-processing (after)

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data, but only
# below Sepal.Length of 7
iris |>
  ggplot(mapping =
           aes(x = Sepal.Length,
             y = Sepal.Width,
             colour = Species)) +
  geom_point() +
  xlim(NA, 7)
```

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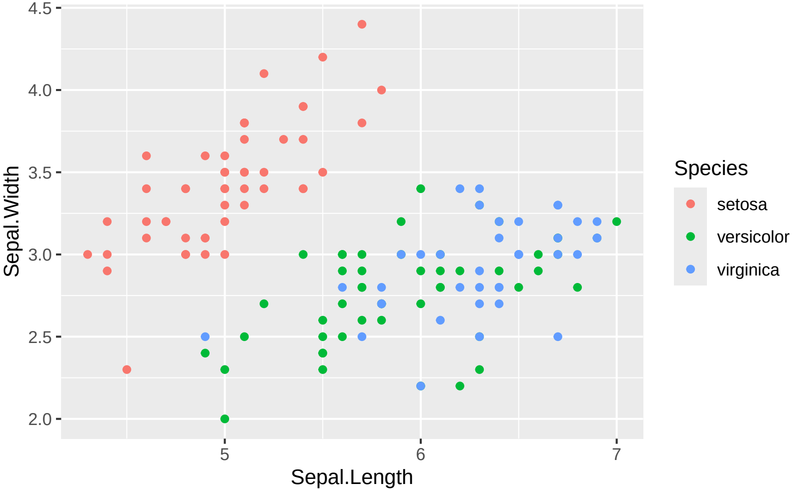
# Scaling through pre-processing (after): Image

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# Pros and Cons of using xlim() and ylim()

- Faster, but not always clear how one may limit data

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- Faster, but not always clear how one may limit data
- May affect some statistics (like means etc.) in graph

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# Pros and Cons of using xlim() and ylim()

- Faster, but not always clear how one may limit data
- May affect some statistics (like means etc.) in graph
- Use coord_cartesian(xlim, ylim) for just “zooming in” while leaving underlying data intact

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# Basic labelling

- Labels are essential for understanding of graph

## Basic labelling

- Labels are essential for understanding of graph
- add labels using + labs(x, y, title, subtitle)

## Basic labelling

- Labels are essential for understanding of graph
- add labels using + labs(x, y, title, subtitle)
- don't forget to enclose labels in " "

# Great example: Labelling


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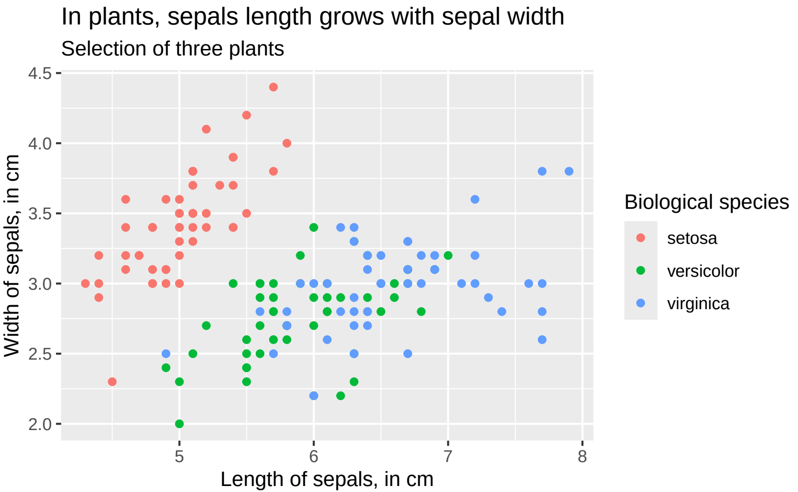
Source: www.ft.com

# labs(): Example code

```
# Load ggplot2
library(ggplot2)
# Scatterplot using iris data set with labs()
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point() +
  labs(
    x = "Length of sepals, in cm",
    y = "Width of sepals, in cm",
    colour = "Biological species",
    title = "In plants, sepals length grows with sepal widt
    subtitle = "Selection of three plants"
```

# labs(): Example image



# Exercise: Create one graph

- Re-create a graph from earlier

# Exercise: Create one graph

- Re-create a graph from earlier
- Limit the x and y variables using xlim() and ylim()

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# Exercise: Create one graph

- Re-create a graph from earlier
- Limit the x and y variables using xlim() and ylim()
- Add labels to the graph

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# Some additional resources

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# Financial Times Visual Vocabulary

**FT Visual Visual Vocabulary**

Source: ft.com/vocabulary

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# The R Graphics Cookbook

- Another great source for learning and troubleshooting

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# The R Graphics Cookbook

- Another great source for learning and troubleshooting
- Has a lot of different standard graphs

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- Has a lot of different standard graphs
- Provides good solutions to standard problems

# The R Graphics Cookbook

- Another great source for learning and troubleshooting
- Has a lot of different standard graphs
- Provides good solutions to standard problems
- https://r-graphics.org/

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# Wrapping up

# Which topics to cover?

- Which topics would you like me to still cover (more)?

# Which topics to cover?

- Which topics would you like me to still cover (more)?
- One suggestions could be Git

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# Working on projects